Exercise Session 10: **Magnetic Fields in Matter 1**

<u>Question 1:</u>

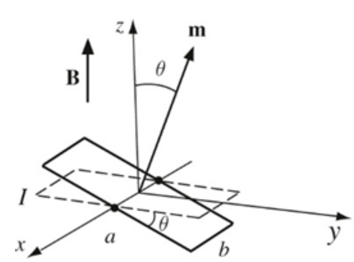
a) In atomic terms, explain the origin of paramagnetism and diamagnetism.
b) Then compare these with ferromagnetism by discussing why ferromagnetic materials can remain strongly magnetized after an external magnetic field is removed, whereas paramagnetic materials usually do not.
c) Finally, explain why diamagnetic and paramagnetic effects are generally much weaker than ferromagnetism.
d) Which of the following materials would you expect to be paramagnetic and which diamagnetic: water, magnesium, silver, carbon, sodium, sodium chloride (NaCl), and molecular oxygen (O_2)? Briefly justify your answer in terms of electronic structure.

<u>Question 2:</u>

A small rectangular current loop of sides $a$and b, carrying a steady current I, is placed in a uniform magnetic field

$$\mathbf{B} = B\hat{\mathbf{z}}.$$

The loop is tilted so that its magnetic dipole moment $\mathbf{m}$ makes an angle θ with the $+z$-axis. Answer the following:



a) Why does the loop experience a torque in this uniform magnetic field but no net force?

b) What does the torque

$$\mathbf{N} = \mathbf{m} \times \mathbf{B}$$

tend to do to the orientation of the magnetic dipole?

c) Even though the forces on opposite sides of the loop cancel in translation, why can they still produce a turning effect?

d) Compare this magnetic torque with the torque on an electric dipole in a uniform electric field. What is the key physical similarity, and what is one important physical difference between the two cases?

Question 3:

A piece of magnetic material occupies some finite region of space. There is no free current flowing through the material, but the material is magnetized, so it has a magnetization vector $\mathbf{M}(\mathbf{r})$. The magnetic field of such an object can be described in terms of equivalent bound currents:

$$\mathbf{J}_b = \nabla \times \mathbf{M}, \mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}}.$$

Answer the following:

a) Why can a magnetized object produce a magnetic field even when no free current is flowing through it?

b) What is the physical meaning of the bound volume current density

$$\mathbf{J}_b = \nabla \times \mathbf{M}?$$

c) What is the physical meaning of the bound surface current density

$$\mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}}?$$

d) Why can a uniformly magnetized object have zero bound volume current but still have a nonzero bound surface current?

e) How does nonuniform magnetization inside a material give rise to internal bound currents?

Question 4:

A long hollow cylindrical conductor has inner radius $a$and outer radius b. It carries a total free current I, uniformly distributed over the conducting cross-section, in the $+z$-direction. The material is weakly diamagnetic.

(a) Use Ampere's law for **H** to find **H** in the regions $s < a$, $a < s < b$, and $s > b$.

(b) State the direction of **H** in each region.

(c) Briefly explain why magnetization does not change the calculation of **H**.